

Text S1: Materials and methods for glasshouse experiments

Soybean growth experiment for architectural data collection

Soybean seeds of wild type parent Bragg and derived mutant *nts1116* (surface-sterilised with 3% H₂O₂ in 70% ethanol for 3 minutes) were sown into autoclaved pots (20 cm in diameter) filled with autoclaved grade 2 vermiculite. The pots were then watered until complete saturation. All plants were kept in an air-conditioned glasshouse with controlled temperature of 28 °C during day and 23 °C at night. On the day after sowing, the seedlings were inoculated with a commercial peat containing *Bradyrhizobium japonicum* strain CB1809 (130 g peat CB1809 mixed with 2,800 mL water). Five plants were kept in each pot after culling on the fifth day. The plants were fertilised twice during this experiment – on the third day and the eighth day after sowing - with B&D nutrient solution [1] plus 2 mM KNO₃. This level of nitrate stimulates plant growth and has minimal effect on nodulation in soybean [2]. The developmental data of shoot and root components, such as internode, cotyledon, unifoliate leaf, trifoliate leaf and leaflets, primary root, lateral root and nodules (including nodule number and nodule distribution) were collected every two days from the third day after sowing by destructive sampling of five plants for each genotype.

Soybean graft experiment

Soybean seeds of Bragg and *nts1116* were surface-sterilised, planted, and grown as described above. Grafting in the hypocotyl or epicotyl was carried out ten days post-sowing and the grafted plants were inoculated with a liquid YMB culture (2 g/L mannitol, 0.4 g/L yeast extract, 0.5 g/L K₂HPO₄, 0.2 g/L MgSO₄ • 7H₂O, 0.1 g/L NaCl, pH 6.8) of CB1809 on the day after grafting. The plants were irrigated with B&D nutrient plus 1 mM KNO₃ solution. The nodule numbers were scored from the grafted plants two weeks after grafting.

References

1. Broughton WJ, Dilworth MJ (1971) Control of leghaemoglobin synthesis in snake beans. *Biochem J* 125: 1075-1080.
2. Carroll BJ, McNeil DL, Gresshoff PM (1985) Isolation and properties of soybean [*Glycine max* (L.) Merr.] mutants that nodulate in the presence of high nitrate concentrations. *P Natl Acad Sci USA* 82: 4162-4166.

Text S2: Growth data collection and model construction

Methods for growth data collection

To collect growth data, the shoot structure was mapped as sequences of symbols [1] where each symbol represents a particular type of plant component (such as internode, cotyledon, petiole and leaflet). For the root system, we classify its components as primary root, first-order lateral roots and nodules. The second-order lateral roots were not taken into account in this work. To capture the branching pattern and help in identifying the first-order lateral roots, we developed an “RULD” root mapping method [2] (Figure 1). We also developed a nodule positioning method to help in recording the distribution information of nodulation (Figure 2).

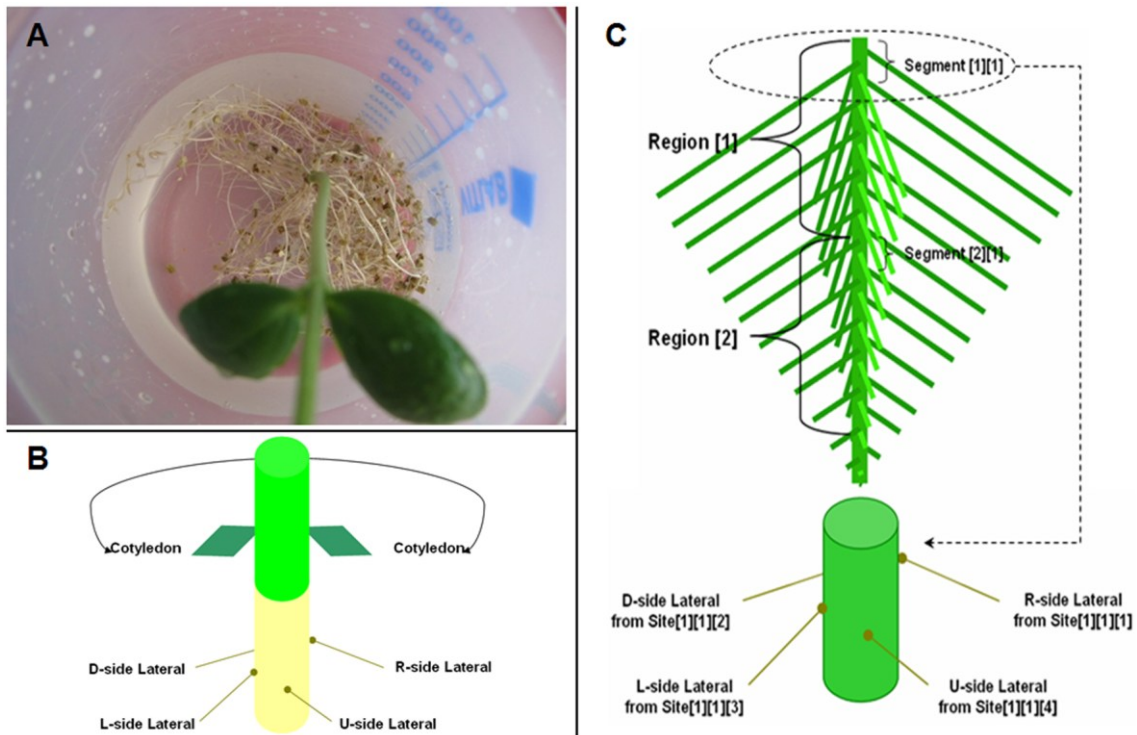


Figure 1. The “RULD” root mapping method. Based on the observed lateral root branching pattern from soybean plants (as shown in image A), the first-order lateral roots were characterised into four sides (R, U, L and D) according to the relative positions of their starting points to the obtuse angle composed by cotyledons in the horizontal plane (as demonstrated in image B). Then the first-order lateral roots were further classified by regions (50 mm long), segments and sites (as illustrated in image C).

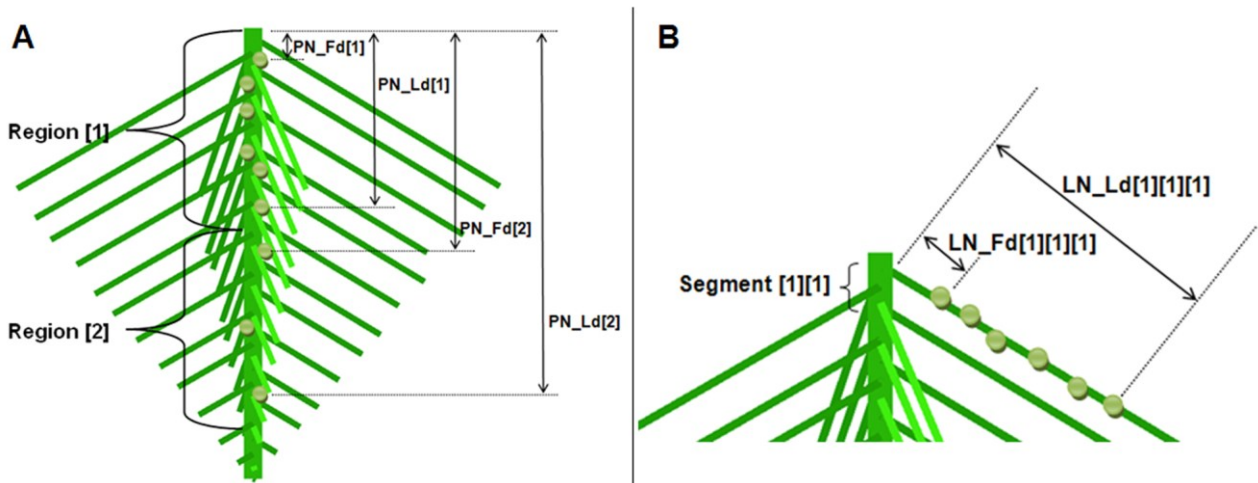


Figure 2. Method for recording distribution information of nodulation. (A) For each region on the primary root (as defined in Figure 1), the distance between the first nodule and the primary-root starting point was measured and recorded as “PN_Fd”; the distance between the last nodule and the primary-root starting point was measured and recorded as “PN_Ld”. (B) For the lateral root nodules, the same method was used with the distances (LN_Fd and LN_Ld) measured from the lateral-root starting point. The positions of the first nodule and the last nodule determine the location of a nodulation section. The nodule density for a section was calculated based on its length and the number of nodules within the section.

Growth data

In analysing the raw data collected with the above methods, some mean values decreased with increasing day, due to the destructive sampling for measurement. The data (except for cotyledon dry weight) were processed to remove this anomaly by not allowing values to decrease, so that normal growth patterns could be simulated. For example, if l_6 and l_8 represent the length of an internode on the 6th day and the 8th day and if l_6 is bigger than l_8 , then the value of l_8 is changed to be the same as l_6 . The processed architectural data are given in Tables 1 - 12. The data on *nts1116* leaf dry weight, needed for building the functional-structural model, are given in Table 13.

Table 1. Dimension of shoot organs (Bragg).

Day			4	6	8	10	12	14	16
Organ Dimension (mm)									
Phytomer 1	Internode	Length	17.3	33.0	65.6	77.4	77.4	77.4	77.4

		Diameter		2.1	2.3	2.6	2.9	2.9	2.9	2.9	
	Cotyledon 1	Length		16.6	20.1	29.6	30.9	31.2	31.2	31.2	
		Width		8.8	12.2	17.2	18.6	18.6	18.6	18.6	
	Cotyledon 2	Length		16.6	20.1	29.8	29.9	30.6	30.6	30.6	
		Width		8.8	12.2	16.8	16.8	18.0	18.0	18.0	
Phytomer 2	Internode	Length		0	0	20.7	54.7	69.2	85.8	95.2	
		Diameter		0	0	1.8	1.9	2.0	2.1	2.4	
	Unifoliate 1	Petiole	Length	0	0.8	5.2	9.8	10.6	11.8	15.5	
			Diameter	0	0	1.1	1.1	1.1	1.3	1.3	
		Surface	Length	0	10.5	12.0	18.9	23.2	26.0	27.0	
			Width	0	4.2	8.2	19.0	26.0	30.6	34.8	
	Unifoliate 2	Petiole	Length	0	0.8	5.1	8.2	12.2	15.2	18.8	
			Diameter	0	0	1.1	1.1	1.1	1.2	1.5	
		Surface	Length	0	10.5	10.5	19.0	27.6	31.2	33.5	
			Width	0	4.2	6.3	18.8	31.2	37.2	44.6	
Phytomer 3	Internode	Length		0	0	0	0	9.6	32.8	52.0	
		Diameter		0	0	0	0	1.7	2.2	2.7	
	Trifoliate	Petiole	Length	0	0	0	0	7.0	26.6	49.2	
			Diameter	0	0	0	0	1.3	2.8	2.8	
		Leaflet 1	Petiolule	Length	0	0	0	0	3.0	4.4	5.7
				Diameter	0	0	0	0	0.9	1.2	1.5
			Surface	Length	0	0	0	10.4	25.8	59.2	80.2
				Width	0	0	0	6.0	14.8	36.0	49.2
		Leaflet 2	Petiolule	Length	0	0	0	0	1.8	2.0	3.1
				Diameter	0	0	0	0	0.8	1.0	1.4
			Surface	Length	0	0	0	0	23.4	51.2	72.7
				Width	0	0	0	0	9.2	30.0	39.3
		Leaflet 3	Petiolule	Length	0	0	0	0	1.8	2.0	3.1
				Diameter	0	0	0	0	0.8	1.0	1.3
			Surface	Length	0	0	0	0	22.0	51.8	75.1
				Width	0	0	0	0	9.0	34.6	40.5
Phytomer 4	Internode	Length		0	0	0	0	0	3.4	13.2	
		Diameter		0	0	0	0	0	1.3	2.1	
	Trifoliate	Petiole		Length	0	0	0	0	0	2.8	15.5
				Diameter	0	0	0	0	0	1.0	1.5
		Leaflet 1	Petiolule	Length	0	0	0	0	0	2.2	5.7
				Diameter	0	0	0	0	0	0.8	1.3
			Surface	Length	0	0	0	0	0	19.0	46.6
				Width	0	0	0	0	0	8.8	27.6
		Leaflet 2	Petiolule	Length	0	0	0	0	0	1.0	3.0
				Diameter	0	0	0	0	0	0.7	1.2
			Surface	Length	0	0	0	0	0	15.6	34.5
				Width	0	0	0	0	0	4.6	20.2
		Leaflet 3	Petiolule	Length	0	0	0	0	0	1.0	2.7
				Diameter	0	0	0	0	0	0.7	1.2
			Surface	Length	0	0	0	0	0	16.6	33.9
				Width	0	0	0	0	0	4.4	20.2

Table 2. Length of primary and lateral roots (Bragg).

Day		4	6	8	10	12	14	16
Root Length (mm)								
Primary root length		88.3	141.2	231.4	293.2	301.7	301.7	303.8
Average lateral root length	Region 1	0	10	73.6	135.9	135.9	135.9	151.7
	Region 2	0	0	18.4	32.5	47.2	53.4	53.4
	Region 3	0	0	7	18.4	37.9	37.9	37.9
	Region 4	0	0	0	16.3	27.3	27.3	34
	Region 5	0	0	0	0	8	20.3	24

Table 3. Number of primary and lateral root nodules (Bragg).

Day		4	6	8	10	12	14	16
Nodule Number								
Region 1	Number of primary root nodules	0	0	2	2	5	5	5
	Number of lateral root nodules (overall)	0	0	0	0	2	19	25
Region 2	Number of primary root nodules	0	0	0	0	2	2	2
	Number of lateral root nodules (overall)	0	0	0	0	0	2	14
Region 3	Number of primary root nodules	0	0	0	0	1	1	1
	Number of lateral root nodules (overall)	0	0	0	0	0	0	2
Region 4	Number of primary root nodules	0	0	0	0	0	0	0
	Number of lateral root nodules (overall)	0	0	0	0	0	0	0
Region 5	Number of primary root nodules	0	0	0	0	0	0	0
	Number of lateral root nodules (overall)	0	0	0	0	0	0	0

Table 4. Data for nodule distribution (Bragg).

Region	Root Type	Position of First Nodule (mm)	Position of Last Nodule (mm)	Nodule Interval (mm)
Region 1	Primary root	8.5	25.4	6.1
	Lateral roots (average)	29.5	87.3	16.5
Region 2	Primary root	67.3	77.1	12.6
	Lateral roots (average)	44.8	52.3	3.8
Region 3	Primary root	114.5	121.1	10.6
	Lateral roots (average)	N/A	N/A	N/A
Region 4	Primary root	N/A	N/A	N/A
	Lateral roots (average)	N/A	N/A	N/A
Region 5	Primary root	N/A	N/A	N/A
	Lateral roots (average)	N/A	N/A	N/A

Table 5. Data for nodule growth (Bragg).

Days from nodule initialisation to appearance	Nodule diameter expansion rate (mm/day)	Maximum nodule diameter (mm)
2	0.19	2.6

Table 6. Data for lateral root distribution (Bragg).

Region	Number of Segments	Lateral Interval (mm)	Probability of Lateral Root Generation			
			R Laterals	U Laterals	L Laterals	D Laterals
Region 1	17	0.7	0.7	0.7	0.5	0.6
Region 2	16	0.8	0.6	0.6	0.6	0.6
Region 3	14	0.9	0.6	0.5	0.5	0.4
Region 4	13	1	0.4	0.3	0.4	0.5
Region 5	12	1	0.4	0.4	0.5	0.2

Table 7. Dimension of shoot organs (*nts1116*).

Organ Dimension (mm)					Day	4	6	8	10	12	14	16	
Phytomer 1	Internode	Length			20.4	42.2	77.8	77.8	77.8	77.8	79.6		
		Diameter			1.7	1.7	2.3	2.3	2.3	2.4	2.4		
	Cotyledon 1	Length			13.6	20.6	25.6	26.8	26.8	26.8	26.8		
		Width			8.8	12.2	15.4	16	16	16.3	16.3		
	Cotyledon 2	Length			13.6	20.2	25.4	26.2	26.2	26.2	26.8		
		Width			8.8	12.6	15.8	16.2	16.2	16.2	16.2		
Phytomer 2	Internode	Length			0	0	21.6	54.4	80.9	83.3	83.3		
		Diameter			0	0	1.4	1.6	1.8	2	2.1		
	Unifoliate 1	Petiole	Length			0	1.8	3.6	7	13	14.3	14.3	
			Diameter			0	0	0.7	0.9	1.1	1.2	1.2	
		Surface	Length			0	11	15.2	26.6	30.9	31	37.6	
			Width			0	4.4	12.4	25	36.6	38	41.2	
	Unifoliate 2	Petiole	Length			0	1.8	3.8	6.8	11.9	19.5	19.5	
			Diameter			0	0	0.8	0.9	1.3	1.4	1.4	
		Surface	Length			0	11	17.6	24.8	32.4	37.3	38.4	
			Width			0	4.4	15	24.6	37.8	38.5	44.4	
Phytomer 3	Internode	Length			0	0	0	0	7	21.7	28.6		
		Diameter			0	0	0	0	1.4	1.8	2.1		
	Trifoliate	Petiole	Length			0	0	0	0	3.9	16.8	23.4	
			Diameter			0	0	0	0	1.1	1.3	1.6	
		Leaflet 1	Petiolule	Length			0	0	0	0	2	5	6.2
				Diameter			0	0	0	0	0.6	1.2	1.3
			Surface	Length			0	0	0	8.2	16.8	42.5	55.4
				Width			0	0	0	2.4	4.9	26	34.4
		Leaflet 2	Petiolule	Length			0	0	0	0	1.6	2.3	2.3
				Diameter			0	0	0	0	0.6	1	1.1
			Surface	Length			0	0	0	0	14.5	37.2	50.6
				Width			0	0	0	0	4	27	28.4
		Leaflet 3	Petiolule	Length			0	0	0	0	1.5	2.5	2.5
				Diameter			0	0	0	0	0.6	1	1.1
			Surface	Length			0	0	0	0	14.3	40.7	51.4
				Width			0	0	0	0	3.9	20.7	29
Phytomer 4	Internode	Length			0	0	0	0	0	3	3.6		
		Diameter			0	0	0	0	0	1.4	1.4		
	Trifoliate	Petiole		Length			0	0	0	0	2.3	3.6	
				Diameter			0	0	0	0	0.8	1	
		Leaflet 1	Petiolule	Length			0	0	0	0	0	1.5	2.8
				Diameter			0	0	0	0	0	0.8	0.9
			Surface	Length			0	0	0	0	0	14.8	20.8
				Width			0	0	0	0	0	4	10.4
		Leaflet 2	Petiolule	Length			0	0	0	0	0	0.8	1.4
				Diameter			0	0	0	0	0	0.7	0.8
			Surface	Length			0	0	0	0	0	11.8	16
				Width			0	0	0	0	0	3	7.2
		Leaflet 3	Petiolule	Length			0	0	0	0	0	1	1.4
				Diameter			0	0	0	0	0	0.6	0.8
			Surface	Length			0	0	0	0	0	11.8	17.2
				Width			0	0	0	0	0	3.2	

Table 8. Length of primary and lateral roots (*nts1116*).

Day		4	6	8	10	12	14	16
Root Length (mm)								
Primary root length		80.8	164.4	214.8	231.5	243	243	253.8
Average lateral root length	Region 1	0	12.9	65.1	98.2	98.2	138.4	141.4
	Region 2	0	0	17.4	24.3	33.5	33.5	33.5
	Region 3	0	0	5.4	14.4	16.1	16.4	28.5
	Region 4	0	0	0	10	32.3	32.3	32.3
	Region 5	0	0	0	2.8	13.1	13.1	20.8

Table 9. Number of primary and lateral root nodules (*nts1116*).

Day		4	6	8	10	12	14	16
Nodule Number								
Region 1	Number of primary root nodules	0	0	7	14	14	14	14
	Number of lateral root nodules (overall)	0	0	0	5	53	74	94
Region 2	Number of primary root nodules	0	0	6	6	8	8	8
	Number of lateral root nodules (overall)	0	0	0	0	2	9	13
Region 3	Number of primary root nodules	0	0	0	1	3	5	5
	Number of lateral root nodules (overall)	0	0	0	0	0	7	7
Region 4	Number of primary root nodules	0	0	0	0	0	0	0
	Number of lateral root nodules (overall)	0	0	0	0	0	0	0
Region 5	Number of primary root nodules	0	0	0	0	0	0	0
	Number of lateral root nodules (overall)	0	0	0	0	0	0	0

Table 10. Data for nodule distribution (*nts1116*).

Region	Root Type	Position of First Nodule (mm)	Position of Last Nodule (mm)	Nodule Interval (mm)
Region 1	Primary root	6.3	43.9	2.5
	Lateral roots (average)	11.2	82.6	6.5
Region 2	Primary root	55.5	88.9	4.5
	Lateral roots (average)	15.5	31.8	3.3
Region 3	Primary root	107.8	124	7.4
	Lateral roots (average)	11.5	18.5	3.5
Region 4	Primary root	N/A	N/A	N/A
	Lateral roots (average)	N/A	N/A	N/A
Region 5	Primary root	N/A	N/A	N/A
	Lateral roots (average)	N/A	N/A	N/A

Table 11. Data for nodule growth (*nts1116*).

Days from nodule initialisation to appearance	Nodule diameter expansion rate (mm/day)	Maximum nodule diameter (mm)
2	0.16	2.6

Table 12. Data for lateral root distribution (*nts1116*).

Region	Number of Segments	Lateral Interval (mm)	Probability of Lateral Root Generation			
			R Laterals	U Laterals	L Laterals	D Laterals
Region 1	16	0.8	0.6	0.6	0.6	0.7
Region 2	13	0.9	0.6	0.5	0.6	0.5
Region 3	12	1.1	0.5	0.6	0.5	0.4
Region 4	13	1	0.4	0.3	0.4	0.4
Region 5	6	2.1	0.4	0.5	0.6	0.3

Table 13. Leaf dry weight (*nts1116*).

Day		4	6	8	10	12	14	16
Leaf	Cotyledon 1	63.9 mg	46.8 mg	41.4 mg	33.9 mg	25.8 mg	24.9 mg	22.8 mg
	Cotyledon 2	63.9 mg	46.8 mg	41.4 mg	33.9 mg	25.8 mg	24.9 mg	22.8 mg
Phytomer 2	Unifoliate 1	0 mg	2.8 mg	7.0 mg	12.4 mg	17.6 mg	17.6 mg	17.6 mg
	Unifoliate 2	0 mg	2.8 mg	7.0 mg	12.4 mg	17.6 mg	17.6 mg	17.6 mg
Phytomer 3	Trifoliate	0 mg	0 mg	0 mg	7.0 mg	23.3 mg	39.7 mg	60.9 mg
Phytomer 4	Trifoliate	0 mg	0 mg	0 mg	0 mg	5.6 mg	11.2 mg	27.9 mg

Methods for model construction

The architectural and functional-structural models were built with the L-system-based language “*cpfg*” in L-studio [3, 4]. In L-system-based architectural models [1, 5, 6], an organ-scale component (such as in internode) is usually simulated with a single module that not only contains its growth information but also represents its structure graphically. In this study, to enable the synchronisation of signalling and developmental processes with different rates and to visualise signal distribution details, we eliminate the graphical role from such modules and only used them to produce a set of standard “sub-modules” for simulation of shoot and root elongation. All sub-modules have the same length, defined as “UNIT” in this work, for both shoot and root structures. For example, for an internode, a leading module “I” is defined to restrict growth (such as elongation, etc.) based on empirical developmental data, while a set of sub-modules “D” play the role of making up structure of the internode. The addition of “D” sub-modules is determined by the empirical elongation data for this internode through the “I” module (Figure 3).

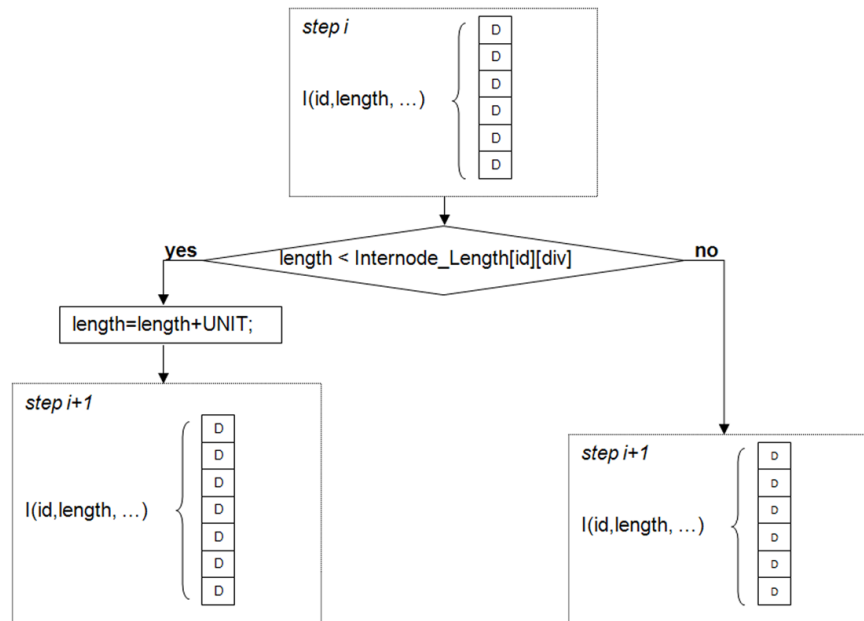


Figure 3. Algorithm with sub-modules for internode elongation. The leading module $I(id, length, \dots)$ carries the basic developmental information of an internode, where “id” is the identification number of this internode and “length” is the current length of this internode. The array “Internode_Length[id][div]” contains the length information of all internodes over time, which is derived from empirical growth data. The “div” is the most basic time unit that can be defined by the user (e.g. it could be an hour or could also be a day). A “div” contains multiple L-system time steps. At a certain step i , if the length of an internode has not reached the maximum length for the current div (say, the value of Internode_Length[id][div] that corresponds to this internode), it will be increased with a standard length “UNIT”, meanwhile a sub-module “D” representing such an increment will be added to existing sub-modules to elongate the internode. Otherwise no addition will be made to the group of sub-modules and the internode length will remain at its previous value. The expressions in this flow chart are in C syntax.

The production of organ-scale components of the shoot is based on a set of developmental rules and relevant empirical data (Figure 4). These production rules are similar to those used in previous L-system-based architectural models [1, 5, 6].

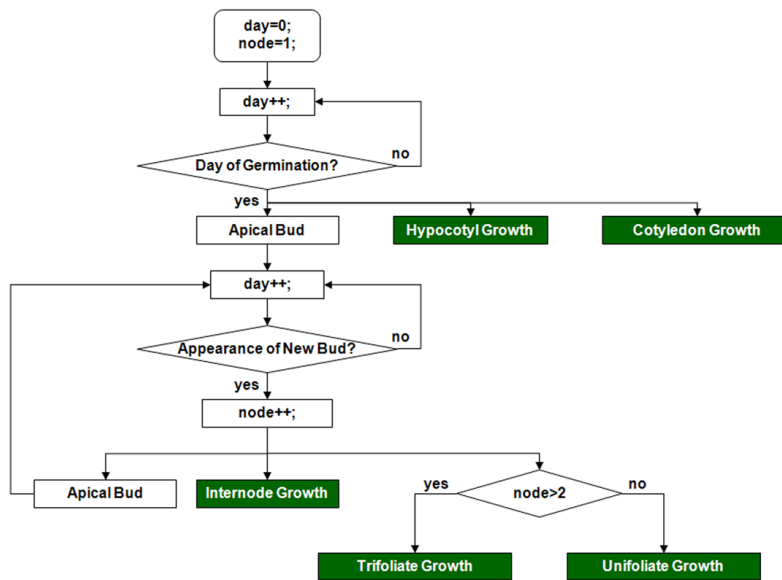


Figure 4. Basic algorithm for production of shoot components. The growth of main stem is led by an apical bud. Once the day of new bud appearance is met, a new node will be added and relevant organ components such as internode and leaf will be produced. The growth of these components (coloured in dark green) behaves individually and simultaneously with the advancement of the main stem. The expressions in this flow chart are in C syntax.

Compared with the clearly “ordered” shoot composition and growth, the development of root system is more complex. To model the elongation of the primary root, we use a module “K” to represent the root tip and to carry the developmental information. When the root tip is heading downwards, a set of sub-modules (with standard length UNIT) is added behind the “K” module to create the primary root structure. Meanwhile, the potential nodulation positions and lateral formation sites are also made available (Figure 5) according to empirical data collected with the root mapping (Figure 1) and nodule recording (Figure 2) methods.

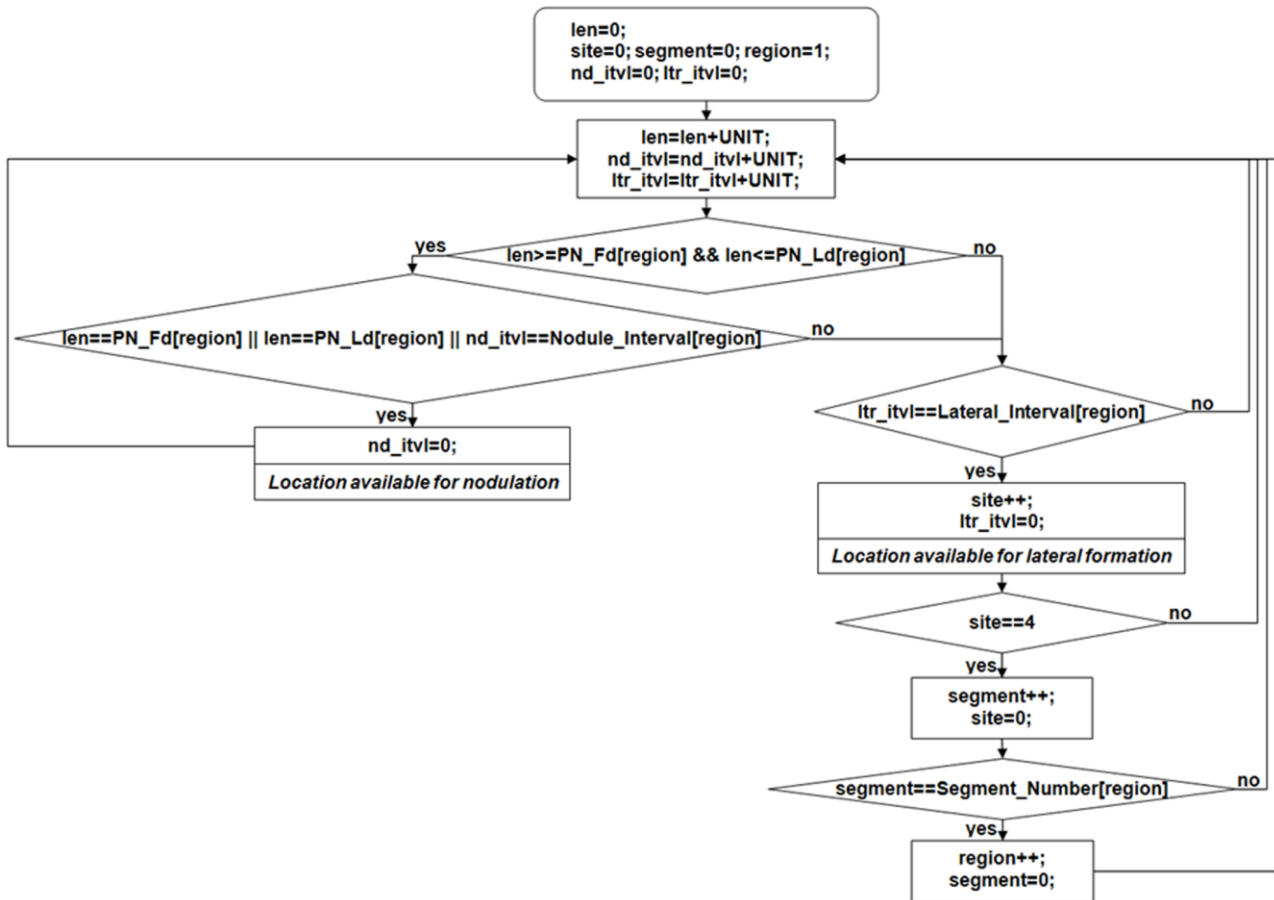


Figure 5. Algorithm for primary root elongation. The variable “len” represents the current length of primary root. The “site”, “segment” and “region” are counters to mark the site, segment and region (Figure 1) where the root tip is currently located. The “nd_itvl” and the “ltr_itvl” are counters to match the interval between two nodules and the interval between two sites. The values of these intervals vary depending on different regions. When the root length reaches the values of PN_Fd[region] or PN_Ld[region], or when it is between PN_Fd[region] and PN_Ld[region] (as shown by Figure 2) and the value of “nd_itvl” reaches the interval between two neighbouring nodules (represented by Nodule_Interval[region]), a location potential for nodule formation will be made available. When the root elongation covers a lateral site interval (represented by Lateral_Interval[region]), a site potential for lateral formation will be made available. When the site number reaches four, a new segment will be started. And when all segments in the current region (represented by Segment_Number[region]) are paved, a new region will be started. The expressions in this flow chart are in C syntax.

In the functional-structural model where the signalling activities are enabled, the signal transport between two neighbouring sub-modules is simulated with context-sensitive L-systems [4]. Compared with traditional context-sensitive L-system modelling where signals are passed from one organ (such as an internode) directly to another, the use of sub-modules (with standard length UNIT) here allows the signals to flow through and/or stop at a certain point within an organ. This improves the accuracy of signal transport, from a spatial point of view. And it also enables the visualisation of signal concentration variation at different locations of the same organ. However, since the developmental events (such as elongation) and the signalling events (such as signal transport) are based on various rates, the capability of signal exchange between two sub-modules was not enough and a coordination mechanism was required to synchronise these multi-rate events. We therefore developed a multi-rate synchronisation algorithm for the functional-structural model (Figure 6). The key role of this algorithm is still played by the standard increment UNIT. In the functional-structural model, the UNIT is not only added to the growing structure as a spatial unit for elongation, but also used with L-system time steps as the basis of system clock (each single step corresponds with a single UNIT). A user-defined constant “DIV” is used in the algorithm to divide a whole day’s time into lower-scale time sections (e.g. 24 hours). Other constants and variables involved in the algorithm can be described as $(E, R, T, C, \tau, \lambda, \theta, u, d, a, step, div, day, counter)$ by Equations 1 - 11.

$$E = \{e_i \mid i = 1, 2, \dots, n\} \quad (1)$$

E is a set of signalling and developmental events (such as signal transport, root elongation, internode elongation and so forth).

$$e_i = \begin{cases} 1 & \text{(activated)} \\ 0 & \text{(stopped)} \end{cases} \quad (2)$$

Each element e_i in E has two alternative values: 1 and 0. When e_i is equal to 1, its corresponding event (event i) is in an “activated” mode and keep going; otherwise the corresponding event is in a “stopped” mode and keep waiting.

$$R = \{r_i \mid i = 1, 2, \dots, n\} \quad (3)$$

Elements in R represent the transport or growth rates (mm/day) of elements in E .

$$\tau : E \rightarrow R \quad (4)$$

The τ means that a one-to-one mapping relationship exists between elements in E and elements in R .

$$T = \{t_i \mid i = 1, 2, \dots, n\} \quad (5)$$

$$u = UNIT \quad (6)$$

$$d = DIV \quad (7)$$

$$\lambda : t_i = \text{ceil}(r_i / (u * d)) \quad (8)$$

Each element t_i in T is the number of time steps that it takes to extend or move through a distance equal to the value of r_i in R ; u is a constant equal to the value of UNIT; d is a constant equal to the value of DIV; the relationship between R and T is represented by λ .

$$a = \max(t_i) \quad (9)$$

The value of a is equal to the maximum value in T .

$$C = \{c_i \mid i = 1, 2, \dots, n\} \quad (10)$$

The set C defines the number of available time steps remaining for advancement of each signalling or

developmental event during a time section (divided by DIV from a day's time).

$$\theta: E \rightarrow C \quad (11)$$

The definition of θ means that a one-to-one mapping relationship exists between elements in E and elements in C .

Variable *step* is used to count the number of L-system time steps conducted. From the start of system running, every group of *a* time steps conducted by the system is counted by a variable *div*. Variable *day* marks the current day. Variable *counter* counts how many L-system time steps are still available for the current time section, initialised with the value of *a*. The flow chart of the algorithm is illustrated by Figure 6.

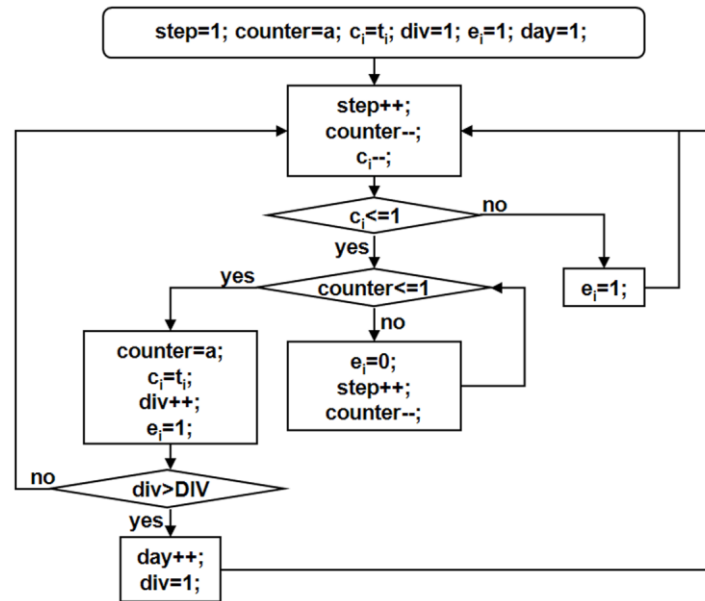


Figure 6. Synchronisation algorithm for coordination of multi-rate developmental and signalling events. The values of “counter” and “*c_i*” are decremented by 1 after the completion of each L-system time step. When “*c_i*” is equal to or lower than 1, its corresponding signalling or development event “*e_i*” will be temporarily stopped until the current time section is finished; otherwise the event “*e_i*” keeps happening. When “counter” is equal to or lower than 1, a new time section will start and the values of “counter” and “*c_i*” will be re-initialised. When the number of time sections (counted by variable “div”) exceeds the value of DIV, a new day will start and the value of “div” will be reinitialised. The expressions in this flow chart are in C syntax.

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Text S3: Assumptions and conditions for virtual experiments

Since our testing target for this first application case is the cotyledon-root and the cotyledon-shoot hypotheses, all other signalling mechanisms, including signal production, transport, perception and function, play a supplementary role to support the running of the AON system. The unknown details of the other signalling mechanisms (excluding the testing target) could be temporarily assumed and manipulated according to

biologists' understanding during virtual experiments but were not tested in this case.

Production of Q

During early stages of soybean nodulation the bacteria still actively produce nod factor, but expanded nodules with mature bacteroids show no stimulation of the NFR1/5 to CCamK cascade needed for nodulation and induction of SDI [1]. Thus, mature nodules have less SDI stimulating activity and we assume the production of Q signal from each nodulation site to be inversely proportional to nodule development stage, strongest at the stage of nodule initialisation but growing weaker as the nodule matures. Since quantitative knowledge about this process is unclear, we used nodule growth potential to represent this inverse relationship:

$$Q_{prdt} = Q_{ini} * (N_{final} - N_{current}) \quad (1)$$

where Q_{prdt} is the quantity of Q signal produced by a nodulation site at a certain moment, Q_{ini} is a parameter used to define the relationship with SDI inhibition threshold (see parameter setting below), N_{final} is the final size of this nodule, and $N_{current}$ is the current size of this nodule.

Signal transport

For the signal transport through roots, stem and petioles, multiple possible patterns of its movement from one tissue to the next are supported by our computational models. For example, the transport could be mass-flow and could also be restricted by certain concentration thresholds. In this case, we assumed mass flow as the transport pattern. The transport rates of Q and SDI were controlled respectively by parameters R_q and R_{sdi} .

Perception of Q and production of SDI

The quantitative pattern of Q perception in a leaf is also supported with multiple options. In this application case, we assumed all Q molecules arriving in the leaf could be fully perceived by *GmNARK*. As a consequent event of this perception, the triggered production of SDI at a certain moment was assumed to be proportional to the perceived Q quantity. Since mRNA expression of *GmNARK* apparently is uniform along the leaf vasculature and the vascular content per leaf is proportional to total leaf biomass [2], we used leaf biomass as the coefficient for this proportion relationship:

$$SDI_{prdt} = Q_{pcpt} * B_{leaf} \quad (2)$$

Where SDI_{prdt} is the quantity of produced SDI signal at a certain moment, Q_{pcpt} is the perceived quantity right before this event and B_{leaf} represents the biomass of this leaf at this moment. At a certain moment, the relationship between SDI_{prdt} and Q_{pcpt} is proportional. However, due to the continuous change of leaf biomass, this process is actually nonlinear over time. For unifoliate and trifoliate leaves, their biomass keeps increasing with plant development, thus their capability to produce SDI also kept increasing in this series of virtual experiments. However, the cotyledon biomass declines as the plant grows, thus the production of SDI from the cotyledons during these virtual experiments kept being weakened.

Function of SDI

When the SDI signal arrives at a potential nodulation site in the root, a threshold is assumed for determining whether nodule initialisation from this site should be inhibited or not. This threshold is defined as SDI_{ihbt} in virtual experiments. If the quantity of SDI signal around a potential nodulation site is higher than SDI_{ihbt} , the potential nodule will be inhibited; otherwise, the potential nodule will be formed.

Parameter setting

The strategy for this application was to adjust parameters for signal production, transport, perception and function within a physiologically appropriate range, as quantitative details about these mechanisms still remain largely unknown. Three qualitative relationships between Q_{ini} and SDI_{ihbt} – namely higher, equal and lower – were represented by setting Q_{ini} at 1, while varying SDI_{ihbt} through values of 0.5, 1 and 2. For signal transport rates, lab experiments demonstrated that auxin, which might play a role in AON, moves at rate of 60 mm/day in soybean (unpublished data), while previous studies suggested that it could be transported “through many plant tissues” at a speed of 240-360 mm/day “in the general direction from the plant’s apex to its roots” [3]. Since SDI might also be other signals and transport rate for Q is unclear, we assumed three values for R_q and R_{sdi} : 60 mm/day, 160 mm/day and 360 mm/day. Combinations allowing different relative speeds for R_q and R_{sdi} gave 27 different conditions for cotyledon-root testing experiments CRH_1 to CRH_27 and cotyledon-shoot experiments CSH_1 to CSH_27, as shown in Table 1.

Table 1. Parameter setting for each virtual experiment.

Experiment ID	Cotyledon Hypothesis	Signal Transport Rates (mm/day)		$Q_{ini} : SDI_{ihbt}$
		Q	SDI	
CRH_1	cotyledon-root	360	360	1
CRH_2	cotyledon-root	360	160	1
CRH_3	cotyledon-root	160	360	1
CRH_4	cotyledon-root	160	160	1
CRH_5	cotyledon-root	360	60	1
CRH_6	cotyledon-root	60	360	1
CRH_7	cotyledon-root	60	60	1
CRH_8	cotyledon-root	160	60	1
CRH_9	cotyledon-root	60	160	1
CRH_10	cotyledon-root	360	360	2
CRH_11	cotyledon-root	360	160	2
CRH_12	cotyledon-root	160	360	2
CRH_13	cotyledon-root	160	160	2
CRH_14	cotyledon-root	360	60	2
CRH_15	cotyledon-root	60	360	2
CRH_16	cotyledon-root	60	60	2
CRH_17	cotyledon-root	160	60	2
CRH_18	cotyledon-root	60	160	2
CRH_19	cotyledon-root	360	360	0.5
CRH_20	cotyledon-root	360	160	0.5
CRH_21	cotyledon-root	160	360	0.5
CRH_22	cotyledon-root	160	160	0.5
CRH_23	cotyledon-root	360	60	0.5
CRH_24	cotyledon-root	60	360	0.5
CRH_25	cotyledon-root	60	60	0.5
CRH_26	cotyledon-root	160	60	0.5
CRH_27	cotyledon-root	60	160	0.5
CSH_1	cotyledon-shoot	360	360	1
CSH_2	cotyledon-shoot	360	160	1
CSH_3	cotyledon-shoot	160	360	1
CSH_4	cotyledon-shoot	160	160	1
CSH_5	cotyledon-shoot	360	60	1
CSH_6	cotyledon-shoot	60	360	1
CSH_7	cotyledon-shoot	60	60	1
CSH_8	cotyledon-shoot	160	60	1
CSH_9	cotyledon-shoot	60	160	1
CSH_10	cotyledon-shoot	360	360	2
CSH_11	cotyledon-shoot	360	160	2
CSH_12	cotyledon-shoot	160	360	2
CSH_13	cotyledon-shoot	160	160	2
CSH_14	cotyledon-shoot	360	60	2
CSH_15	cotyledon-shoot	60	360	2
CSH_16	cotyledon-shoot	60	60	2
CSH_17	cotyledon-shoot	160	60	2
CSH_18	cotyledon-shoot	60	160	2
CSH_19	cotyledon-shoot	360	360	0.5
CSH_20	cotyledon-shoot	360	160	0.5
CSH_21	cotyledon-shoot	160	360	0.5
CSH_22	cotyledon-shoot	160	160	0.5
CSH_23	cotyledon-shoot	360	60	0.5
CSH_24	cotyledon-shoot	60	360	0.5
CSH_25	cotyledon-shoot	60	60	0.5
CSH_26	cotyledon-shoot	160	60	0.5
CSH_27	cotyledon-shoot	60	160	0.5

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